
SDeltaPhi-35 - Minimal Closure Criteria for AGI: A Delta-phi-Based Judgment Framework (v1.2)

With a March 2026 GPT-family mapping example and anticipated objections.

Abstract. This working paper proposes a closure-based judgment framework for Artificial General Intelligence within the SDeltaPhi Formalism. Rather than treating AGI as a vague destination, a matter of human equivalence, or a benchmark label, the paper reformulates AGI as a structured closure problem. Intelligence is treated not as a static possession but as a phase-stabilized capacity for maintaining coherent transition across heterogeneous tasks, environments, and recursive updates. On this basis, AGI is evaluated not by appearance or rhetoric, but by the degree to which a minimal set of operational conditions has become closed.

1. Introduction

AGI discourse often oscillates between spectacle and definitional dispute. In one register, AGI is a promised destination; in another, it is treated as a threshold of human equivalence; in still another, it is reduced to benchmark aggregation. All three frames are unstable. They are rhetorically strong, but operationally weak.

Within SDeltaPhi, AGI is not treated as a symbolic destination. It is treated as a judgment problem. The relevant question is not whether a system appears broadly intelligent, but whether a sufficient set of operational conditions has become closed.

This distinction matters because a system may display striking competence in narrow or semi-general tasks while still failing to preserve stability across heterogeneous environments, long-horizon goals, recursive updates, or world correspondence. In such a case, the system may be powerful, impressive, or transformative, but it should not yet be judged as AGI in the strong sense.

2. Why Definitions Are Insufficient

Human-equivalence definitions are unstable because "human level" is neither singular nor fixed. Benchmark aggregation is useful but incomplete because peak scores do not by themselves prove closure. Impressionistic generality is even weaker: fluency, surprise, and public hype are not criteria.

SDeltaPhi therefore does not begin with a label and seek supporting examples. It begins with closure conditions and asks whether the label is warranted. The issue is not what AGI is called, but which conditions have actually closed.

3. Reframing AGI in the SDeltaPhi Formalism

Within SDeltaPhi, intelligence is not treated as a static property or essence. It is treated as a structured capacity for transition under heterogeneous conditions.

Minimal coordinate language: $\phi(t)$ is the state of a system at time t ; $\Delta\phi$ is state change; G is the transition kernel governing how state evolves; closure is the stabilization of an operational condition such that it persists across task variation, recursive update, and environmental demand without collapse.

On this basis, AGI is not understood as a title granted at the moment of high performance. It is understood as a condition in which multiple operational closures have become sufficiently stable at once.

4. The Five Minimal Closure Criteria

4.1 Global Generality

Definition. Global generality refers to a system's capacity to sustain competent transition across heterogeneous task domains without catastrophic jaggedness or domain-specific collapse.

Closure condition. For diverse environments e_i , $\text{Performance}(G \mid e_i)$ remains above threshold and variance of $\Delta\phi$ across tasks stays below epsilon.

Failure mode if open. The system is locally strong but globally unstable: brilliant in some regions, fragile in others.

4.2 Long-Horizon Recursive Agency

Definition. Long-horizon recursive agency is the ability to preserve and update goal-directed structure across extended sequences rather than merely producing competent local responses.

Closure condition. $\text{Goal}(t)$ remains coherent through $\text{Goal}(t+k)$, while the cost of path interruption exceeds the cost of path continuation.

Failure mode if open. The system appears agentic only in short bursts. It can begin but not sustain.

4.3 World-Model Correspondence

Definition. World-model correspondence is the degree to which internal transition structure remains aligned with external world constraints rather than drifting into merely internally coherent but externally false trajectories.

Closure condition. The distance between $\Delta\phi_{\text{model}}$ and $\Delta\phi_{\text{world}}$ remains sufficiently small under task-relevant stress.

Failure mode if open. The system remains vulnerable to hallucination, brittle abstraction, and plans that collapse in the world.

4.4 Self-Updating Robustness

Definition. Self-updating robustness is the ability to incorporate new information and constraints without catastrophic degradation of structural coherence.

Closure condition. Meaningful self-update occurs, pattern persistence remains above a minimum threshold, and self-damage remains below an instability bound.

Failure mode if open. The system either changes and degrades or preserves itself by refusing genuine adaptation.

4.5 Creative Path Generation

Definition. Creative path generation is not stylistic novelty but the ability to generate, validate, and stabilize genuinely new operational paths not reducible to trivial replay.

Closure condition. A new candidate path G_{prime} is generated, shown to exceed trivial restatement, validated for explanatory or operational value, and stabilized for reuse.

Failure mode if open. The system may look creative while remaining structurally derivative.

5. Open and Closed Conditions

One of the central errors in AGI discourse is the assumption that AGI must be judged as a binary state. Within SDeltaPhi , closure is often uneven. A system may exhibit strong global generality while remaining weak in self-updating robustness. It may show partial agency while still lacking adequate world-model correspondence. The proper question is therefore: which conditions are closed, which remain open, and how stable is the closure pattern across recursive demand?

6. Why AGI Must Be Judged by Closure, Not Impression

A system can look general without being stable. It can plan briefly without sustaining agency. It can sound grounded while misaligning with the world. It can appear creative while remaining derivative. SDeltaPhi therefore rejects impression-based AGI judgments. AGI-like appearance is not AGI closure.

7. Mapping Example: The GPT Family as of March 2026

The closure framework is most useful when applied to real systems. The following example is intentionally provisional. It does not issue a final verdict on "whether GPT is AGI." Instead, it asks which closures appear relatively advanced in the GPT family as of March 2026, which remain partial, and which are still structurally open.

Criterion	March 2026 GPT-family status	Illustrative basis
Global generality	Relatively advanced / near-closure among the five	Official GPT-5 and GPT-5.2 materials describe strong performance across reasoning, coding, multimodal understanding, long context, and professional knowledge work.
Long-horizon recursive agency	Partially closed	Operator and agent-mode integrations, plus GPT-5.2-Codex and GPT-5.3-Codex, indicate stronger tool use and longer task continuation, but not full autonomous long-horizon stability.
World-model correspondence	Intermediate / still open	The family is stronger in multimodal and tool-grounded tasks, but public evidence still centers on text-heavy and agentic workflows rather than robust embodied world correspondence.
Self-updating robustness	Largely open	OpenAI release materials show versioned model improvement and adjustable reasoning depth, but not continuous self-updating learning with stable structural preservation during deployment.
Creative path generation	Strong moments, but not stable closure	The family shows powerful problem-solving and nontrivial synthesis, yet stable generation, validation, and fixation of genuinely new paths remains unclosed.

- Range note: this example uses public OpenAI material and therefore remains conservative. It tracks externally documented capability claims rather than hidden internal metrics.
- Interpretive rule: apparent AGI-like performance is not equivalent to closure. A criterion can look strong in demonstrations while remaining open under heterogeneous stress.
- Practical implication: the GPT family looks further along the path in breadth and tool-mediated continuation than in world-grounded closure, continuous self-updating robustness, or stable creative path fixation.

8. Anticipated Objections

Objection 1. "This is just another AGI definition in disguise."

Reply: The framework does not settle AGI by stipulative definition. It treats 'AGI' as a label whose legitimacy depends on whether specific operational conditions have closed. It is evaluative rather than declarative.

Objection 2. "Closure is only a new word for capability."

Reply: Closure is not a synonym for competence. It denotes stability under heterogeneous stress: persistence across task shifts, recursive updates, and environmental constraints without collapse. The term separates peak performance from phase-stabilized capacity.

Objection 3. "Why these five criteria?"

Reply: The five criteria are proposed as minimal, not exhaustive. They were selected because they correspond to distinct failure modes that repeatedly block general closure: jagged collapse, short-horizon breakdown, world-model divergence, instability under update, and inability to generate and stabilize new paths.

Objection 4. "The framework is still weak on measurement."

Reply: This paper is a minimal criterion framework, not a completed measurement protocol. Its purpose is to define the axes along which measurement must occur and to distinguish apparent capability from closure. Instrumentation is the next layer, not the present one.

Objection 5. "The GPT mapping is interpretive and subjective."

Reply: The mapping is intentionally presented as a worked example rather than a final verdict. Its value lies in localizing disagreement to specific criteria and failure modes, replacing rhetorical dispute with structured contestability.

9. Implications

The shift from definition to closure has several implications. It changes how progress is measured. It changes how governance and capability evaluation should be linked. It also clarifies the practical importance of partially closed systems, which may be neither mere tools nor full AGI yet can still have major real-world impact.

10. Conclusion

This paper argues that AGI should not be treated as a definitional object but as a closure judgment problem within the SDeltaPhi Formalism.

The point is not to reject AGI discourse, but to reframe it. Human equivalence, benchmark aggregation, and intuitive impressions of generality are not useless, but they are insufficient. They describe appearances and comparisons; they do not determine closure.

Within SDeltaPhi, AGI is not the moment at which a system appears generally intelligent. It is the point at which a sufficient set of operational conditions becomes closed. The proper question is therefore not 'What is AGI?' but 'Which operational conditions have closed, which remain open, and how stable is that closure under recursive demand?'

References

OpenAI. "Introducing GPT-5." 7 Aug 2025. <https://openai.com/index/introducing-gpt-5/> .
OpenAI. "Introducing GPT-5.2." 11 Dec 2025. <https://openai.com/index/introducing-gpt-5-2/> .
OpenAI. "Introducing GPT-5.2-Codex." 18 Dec 2025. <https://openai.com/index/introducing-gpt-5-2-codex/> .
OpenAI. "Introducing Operator." 23 Jan 2025; updated 17 Jul 2025. <https://openai.com/index/introducing-operator/> .
OpenAI Help Center. "Model Release Notes." Accessed Mar 2026.
<https://help.openai.com/en/articles/9624314-model-release-notes> .